01 / Start with a question



Watercolour study 1 / contextual illustration, not measurement evidence

THE OBSERVATION

Notice before you measure.

This field journal explores quantities, units and written observations through a familiar kitchen setting. Its numbers are invented learning examples, not readings from the pictured instruments. Start by saying what you want to describe.

MODEL / METHOD NOTE

Observation → quantity → unit → method.

TRY / REFLECT

Name one quantity you could describe without claiming a recipe or safety outcome.

32 PAGES / 16 INDEPENDENT MAIN ILLUSTRATIONS · MODEL NUMBERS, NOT REAL READINGS

01 / Start with a question



Watercolour study 1 / contextual illustration, not measurement evidence

THE OBSERVATION

Make the question small.

“How much is in this bowl?” is ambiguous. It might ask for mass, volume, price or a portion count. A useful measurement question names one quantity, the object and the intended comparison.

MODEL / METHOD NOTE

Example question: What is the mass of these dry oats, excluding the bowl?

TRY / REFLECT

Rewrite a broad kitchen question so a second reader knows what would be measured.

01 / Start with a question



Watercolour study 2 / contextual illustration, not measurement evidence

THE OBSERVATION

A tool does not define the claim.

A scale and a jug answer different questions. The illustration is contextual, not a calibrated instrument. Select a method appropriate to the quantity and describe the method rather than inferring accuracy from an attractive picture.

MODEL / METHOD NOTE

Mass and volume are different quantities.

TRY / REFLECT

Which question calls for a mass reading, and which calls for a volume reading?

01 / Start with a question



Watercolour study 2 / contextual illustration, not measurement evidence

THE OBSERVATION

Attach the unit immediately.

A bare number is not enough. “200” might mean grams, millilitres or a count. Keep the quantity and unit together in notes and handoffs so someone else does not have to reconstruct the meaning.

MODEL / METHOD NOTE

200 g is a mass expression; 200 mL is a volume expression.

TRY / REFLECT

Write two complete observations using different quantity-unit pairs.

02 / Read the units



Watercolour study 3 / contextual illustration, not measurement evidence

THE OBSERVATION

One thousand grams.

For this unit relationship, one kilogram equals one thousand grams. Converting the written unit does not change the physical quantity, improve the instrument or make an observation more accurate.

MODEL / METHOD NOTE

1 kg = 1,000 g; 0.25 kg = 250 g.

TRY / REFLECT

Express 0.6 kg in grams without changing the quantity described.

02 / Read the units



Watercolour study 3 / contextual illustration, not measurement evidence

THE OBSERVATION

One thousand millilitres.

One litre equals one thousand millilitres. Keep this volume conversion separate from mass. A conversion between litres and grams needs additional material information and cannot be inferred from the unit names alone.

MODEL / METHOD NOTE

1 L = 1,000 mL; 0.4 L = 400 mL.

TRY / REFLECT

Explain why 400 mL cannot automatically be renamed 400 g for every material.

02 / Read the units



Watercolour study 4 / contextual illustration, not measurement evidence

THE OBSERVATION

The container belongs in the method.

If a reading includes a bowl, that reading describes bowl plus contents. Record whether the container was excluded by the method or subtracted later. Do not silently compare a combined reading with a contents-only reading.

MODEL / METHOD NOTE

Name the empty-container step before recording a contents-only result.

TRY / REFLECT

What would be missing from a note that says only “the bowl reads 320 g”?

02 / Read the units



Watercolour study 4 / contextual illustration, not measurement evidence

THE OBSERVATION

Subtract like quantities.

In this invented arithmetic example, bowl plus contents is 320 g and the empty bowl is 120 g. Subtracting the container gives 200 g for the contents within the example. This is not evidence of scale accuracy.

MODEL / METHOD NOTE

320 g − 120 g = 200 g.

TRY / REFLECT

If the combined example is 410 g and the bowl is 160 g, what mass remains?

03 / Keep a method



Watercolour study 5 / contextual illustration, not measurement evidence

THE OBSERVATION

Describe the fill.

A cup of dry material may be loosely filled, compressed or heaped. Those are different preparation methods. Name the chosen method before comparing results; the drawing is an illustration, not a universally specified kitchen standard.

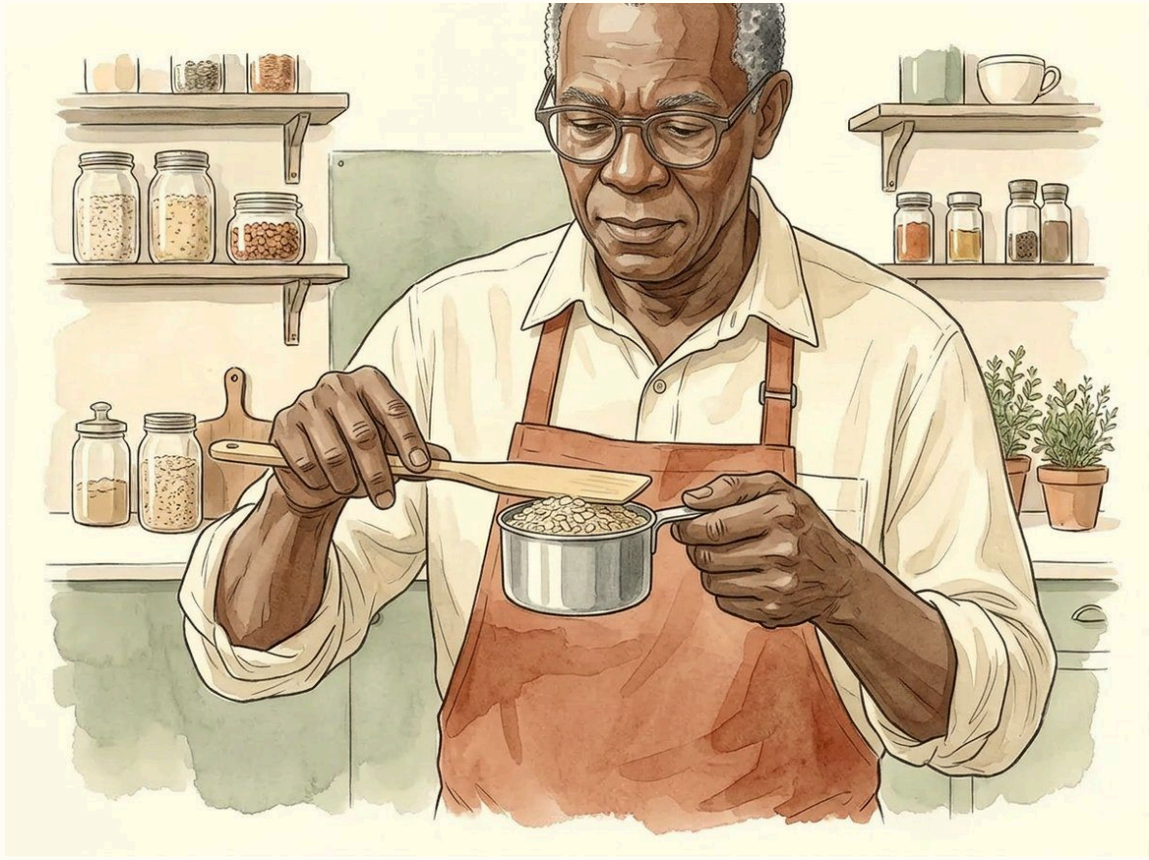
MODEL / METHOD NOTE

Record: material / container / fill method / unit.

TRY / REFLECT

Which part of a fill method would you need to repeat for a fair comparison?

03 / Keep a method



Watercolour study 5 / contextual illustration, not measurement evidence

THE OBSERVATION

Volume is not a material shortcut.

Equal volumes of different materials need not have equal mass. A cup of oats and a cup of another material cannot be exchanged by mass without a justified relationship. This journal supplies no density table or recipe conversion.

MODEL / METHOD NOTE

Same stated volume does not guarantee the same mass.

TRY / REFLECT

Identify the unsupported assumption in “every full cup weighs the same”.

03 / Keep a method



Watercolour study 6 / contextual illustration, not measurement evidence

THE OBSERVATION

Name how you read a level.

A liquid-level observation depends on the vessel and reading method. Describe the position and tool used, and follow the actual tool instructions. The pictured jug has no calibrated markings and should not be used to estimate a numeric reading.

MODEL / METHOD NOTE

Contextual illustration only; no numeric reading can be taken from the art.

TRY / REFLECT

What method details should accompany a real liquid-volume observation?

03 / Keep a method



Watercolour study 6 / contextual illustration, not measurement evidence

THE OBSERVATION

Hold the comparison conditions.

When comparing two observations, keep the named method and object consistent where the question requires it. If the material, container or preparation changes, record that change rather than presenting the values as directly interchangeable.

MODEL / METHOD NOTE

A changed method belongs beside the changed value.

TRY / REFLECT

Name one change that would prevent a simple like-for-like comparison.

04 / Keep a record



Watercolour study 7 / contextual illustration, not measurement evidence

THE OBSERVATION

Write while the meaning is clear.

A compact note can preserve the quantity, unit, object, method and source. You do not need personal details to describe a kitchen arithmetic example. Write enough for another reader to understand what the value means.

MODEL / METHOD NOTE

Record template: object / quantity / unit / method / context.

TRY / REFLECT

Draft a short observation note without including anyone’s personal information.

04 / Keep a record



Watercolour study 7 / contextual illustration, not measurement evidence

THE OBSERVATION

Digits are not a quality certificate.

A longer number may appear precise, but its appearance does not prove accuracy. Avoid adding unsupported decimal places to a result or suggesting that a finely drawn scale is a calibration record.

MODEL / METHOD NOTE

More displayed digits do not by themselves establish accuracy.

TRY / REFLECT

Why is writing 200.000 g not an automatic improvement over 200 g?

04 / Keep a record



Watercolour study 8 / contextual illustration, not measurement evidence

THE OBSERVATION

Repeat the stated question.

Repeated observations are useful only when their context is recorded. State whether the same material and method were used. Three values are not enough to certify an instrument or establish a general performance claim.

MODEL / METHOD NOTE

Repeated examples should keep labels and method notes attached.

TRY / REFLECT

What context would you preserve beside three repeated readings?

04 / Keep a record



Watercolour study 8 / contextual illustration, not measurement evidence

THE OBSERVATION

An average describes this set.

For the invented values 100 g, 102 g and 101 g, the arithmetic mean is 101 g. That summary describes these three numbers only. It does not establish the true mass or justify a claim about instrument accuracy.

MODEL / METHOD NOTE

(100 + 102 + 101) g ÷ 3 = 101 g.

TRY / REFLECT

Name one question the mean does not answer.

05 / Compare with care



Watercolour study 9 / contextual illustration, not measurement evidence

THE OBSERVATION

A cup needs a definition.

The word cup can refer to different conventions or household vessels. In this model only, cup A is defined as 240 mL and cup B as 250 mL. These definitions explain the arithmetic; they are not a universal rule for every cup.

MODEL / METHOD NOTE

Model definitions: A = 240 mL; B = 250 mL; difference = 10 mL.

TRY / REFLECT

Why must the chosen cup definition stay in a shared note?

05 / Compare with care



Watercolour study 9 / contextual illustration, not measurement evidence

THE OBSERVATION

Convert only a stated relationship.

A written conversion needs a known relationship between units of the same quantity, or additional justified information for unlike quantities. Do not fill a gap with a guess just to produce a clean-looking answer.

MODEL / METHOD NOTE

Known: 1 L = 1,000 mL. Not supplied: material density.

TRY / REFLECT

Which conversion can you complete from this page, and which must remain unresolved?

05 / Compare with care



Watercolour study 10 / contextual illustration, not measurement evidence

THE OBSERVATION

Double the named amount.

In this arithmetic model, doubling a specified 125 g amount gives 250 g. The operation describes a quantity relationship, not a recommendation to scale a recipe, equipment setting or process time.

MODEL / METHOD NOTE

$2 \times 125\ \text{g} = 250\ \text{g}$.

TRY / REFLECT

Double 175 g and retain the unit in the answer.

05 / Compare with care



Watercolour study 10 / contextual illustration, not measurement evidence

THE OBSERVATION

Halve without losing the unit.

Half of the model amount 180 g is 90 g. Keeping the unit makes the result readable outside this page. A numerical scaling example does not imply that every real cooking process can be scaled in the same way.

MODEL / METHOD NOTE

180 g ÷ 2 = 90 g.

TRY / REFLECT

Explain what this arithmetic does—and does not—say about a real recipe.

06 / Describe the model



Watercolour study 11 / contextual illustration, not measurement evidence

THE OBSERVATION

Equal groups are an assumption.

If a model explicitly gives three equal groups of 200 g, their combined mass is 600 g. Equal grouping is a stated assumption here, not a conclusion drawn from an illustration of several bowls.

MODEL / METHOD NOTE

3 × 200 g = 600 g, assuming three equal model groups.

TRY / REFLECT

What would change if the groups were not equal?

06 / Describe the model



Watercolour study 11 / contextual illustration, not measurement evidence

THE OBSERVATION

Separate fixed and variable parts.

A container contribution and a contents contribution play different roles in a combined reading. If contents change while the same container is retained, record which part was held fixed. Do not compare totals without this context.

MODEL / METHOD NOTE

Combined model = container contribution + contents contribution.

TRY / REFLECT

Which part would stay fixed if only the contents changed?

06 / Describe the model



Watercolour study 12 / contextual illustration, not measurement evidence

THE OBSERVATION

A discrepancy is a question.

Different numbers can arise from different objects, methods, notation or instruments. A discrepancy alone does not prove which explanation is correct. Keep the original records and list a checkable question before assigning a cause.

MODEL / METHOD NOTE

Difference observed; cause not established.

TRY / REFLECT

Write a question that would help distinguish a notation change from an object change.

06 / Describe the model



Watercolour study 12 / contextual illustration, not measurement evidence

THE OBSERVATION

A range is a description.

For the invented readings 98 g, 100 g and 102 g, the range is 4 g. This is the difference between largest and smallest in this set, not an uncertainty guarantee or a limit on future readings.

MODEL / METHOD NOTE

102 g − 98 g = 4 g.

TRY / REFLECT

Why could a later observation lie outside this three-value range?

07 / Share the context



Watercolour study 13 / contextual illustration, not measurement evidence

THE OBSERVATION

Keep the origin attached.

When sharing an example, state whether the values were invented, observed or copied from a named source. All numbers in this journal are learning models. The illustrations do not document an actual measurement session.

MODEL / METHOD NOTE

Source label: fictional arithmetic for measurement literacy.

TRY / REFLECT

How would you stop someone mistaking this page for real measurement evidence?

07 / Share the context



Watercolour study 13 / contextual illustration, not measurement evidence

THE OBSERVATION

Missing is not zero.

An unrecorded value means the record does not supply a value. Replacing it with zero introduces information that was never observed. Keep a missing state explicit until there is an appropriate source or method to resolve it.

MODEL / METHOD NOTE

Not recorded ≠ observed zero.

TRY / REFLECT

Describe a missing reading without inventing a replacement number.

07 / Share the context



Watercolour study 14 / contextual illustration, not measurement evidence

THE OBSERVATION

Leave unsupported certainty out.

These examples do not quantify instrument uncertainty or certify a measurement process. If a real decision requires such information, seek an appropriate documented method and qualified support instead of inferring it from these pages.

MODEL / METHOD NOTE

No calibration, uncertainty bound or safety-critical suitability is claimed.

TRY / REFLECT

Which claim would exceed the evidence supplied by this notebook?

07 / Share the context



Watercolour study 14 / contextual illustration, not measurement evidence

THE OBSERVATION

Hand over a method, not a guess.

A useful handoff gives the question, the named quantity and unit, the method and known limitations. Ask the next reader to explain the note back in their own words so unclear terms can be corrected.

MODEL / METHOD NOTE

Question → quantity/unit → method → limitations → next reader.

TRY / REFLECT

Which term would you clarify before a second person repeats the example?

08 / Close the notebook



Watercolour study 15 / contextual illustration, not measurement evidence

THE OBSERVATION

Revise without erasing the trail.

If a notation error is found, preserve the distinction between the original note and the corrected version. A correction should explain what changed; it is not a reason to invent a missing observation or conceal a method difference.

MODEL / METHOD NOTE

Correction example: unit label clarified; original observation not re-created.

TRY / REFLECT

How would you label a correction without pretending to have remeasured the object?

08 / Close the notebook



Watercolour study 15 / contextual illustration, not measurement evidence

THE OBSERVATION

Check the unit in the answer.

Review arithmetic and units together. Adding 100 g to 50 g gives 150 g within the model. Adding a mass to an unrelated volume is not justified just because both values are numbers.

MODEL / METHOD NOTE

100 g + 50 g = 150 g; grams + millilitres needs a different explanation.

TRY / REFLECT

Find one valid calculation and one unsupported combination in your own draft notes.

08 / Close the notebook



Watercolour study 16 / contextual illustration, not measurement evidence

THE OBSERVATION

Review the claim before sharing.

Before exporting a note, ask whether a reader can distinguish model arithmetic from real evidence. Review units, assumptions and limitations. A local review mark is a personal reminder, not formal approval or certification.

MODEL / METHOD NOTE

Review checklist: question / unit / method / source / limitation.

TRY / REFLECT

Which limitation must remain visible when this notebook is shared?

08 / Close the notebook



Watercolour study 16 / contextual illustration, not measurement evidence

THE OBSERVATION

Leave a clearer next question.

A useful notebook ends with a question that can be understood and examined, not a claim that every uncertainty has disappeared. Keep the object, quantity, unit and method in view as you plan the next observation.

MODEL / METHOD NOTE

Notice carefully. Name the quantity. Keep the context.

TRY / REFLECT

Write the next small question you would want a reader to understand.